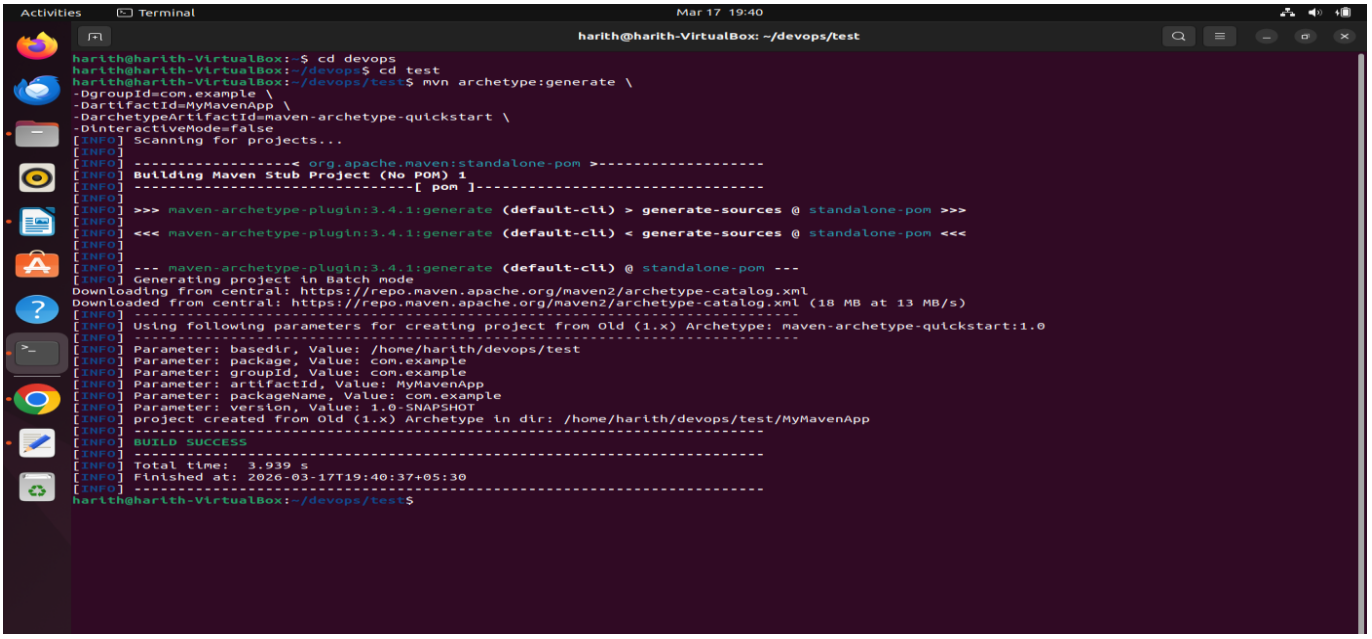
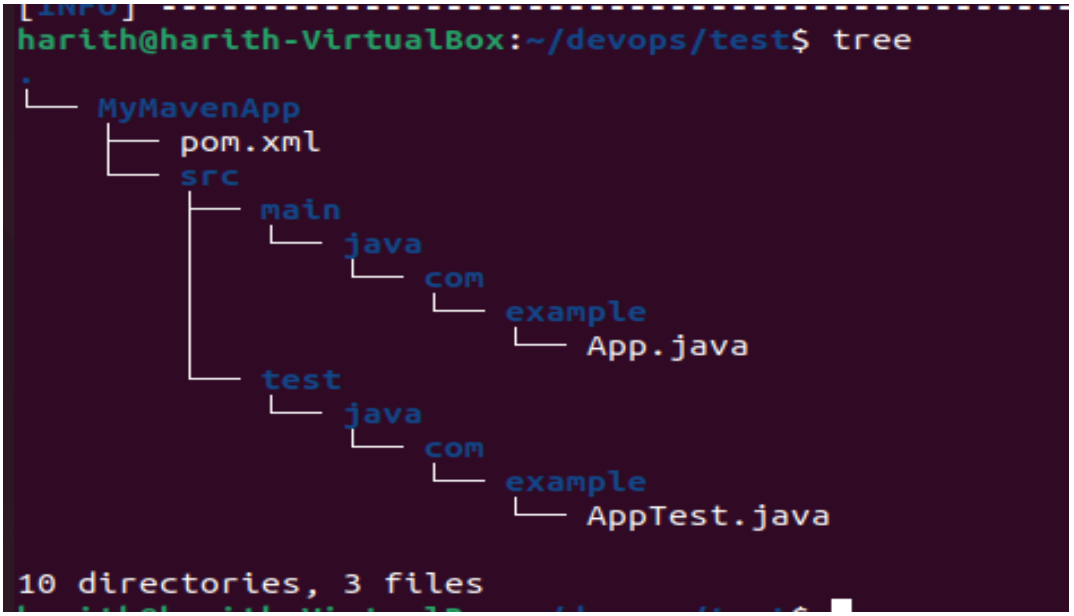
Program 1:Creating a Maven Project

Step 1: Create a Maven project:

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false



This command generates a basic Maven project structure with a sample App.java and AppTest.java.



Step 2: Go to the project directory using command cd MyMavenApp

Step 3: Edit pom.xml: gedit pom.xml

- Go to mvnrepository.com
- Search “junit”
- Copy dependency
- Paste inside <dependenciss>

Open pom.xml ~/devops/test/MyMavenApp Save

App.java pom.xml

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenApp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Bracket match found on line: 12 XML Tab Width: 8 Ln 12, Col 31 INS

Update App.java:

Open App.java ~/devops/test/MyMavenApp/src/main/java/com/example Save

```java
package com.example;

/**
 * Hello world!
 *
 */
public class App
{
    public static void main( String[] args )
    {
        System.out.println( "Hello World!" );
    }
}
```

Step 4: Build the project using command mvn compile

The project building will start like this.

```
harith@harith-VirtualBox:~/devops/test/MyMavenApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:MyMavenApp >-----------------------
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is
[INFO] skip non existing resourceDirectory /home/harith/devops/test/MyMavenApp/src/main/reso
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platf
[INFO] Compiling 1 source file to /home/harith/devops/test/MyMavenApp/target/classes
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.097 s
[INFO] Finished at: 2026-03-17T19:59:38+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenApp$
```

The “Build Success” message ensures that our Maven project is built successfully.

Step 5: Package the project using command mvn package.

Step 6: Run the test on project using command mvn test.

Step 7: To run the Maven application

java -jar MyMavenApp-1.0-SNAPSHOT.jar